Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

a. Team name: Team Atlantis
b. Team leader name: Anand Munjuluri
c. Problem Statement: Manufacturing - Intelligent Operations & Industrial Efficiency

Brief about the idea

Plants already have maintenance software, but it is **rarely kept up to date, because every breakdown has to be typed in by hand**. So when a machine stops, the manager takes time, gathers everything manually and guides the workers by hand, while every hour can cost about $125,000.

PROBLEM AREAS

1. **Every fix typed in by hand**
2. **Manager gathers everything by hand**
3. **Workers guided by word of mouth**
4. **No estimate of repair time**
5. **Spare parts in stock not known upfront**

Our Idea: PlantMemory

Make the first step easy, and the rest gets easier: report a breakdown in seconds, and PlantMemory gathers, drafts and plans the rest.

- **Report it any way** – speak, type or photograph the logbook, in English, Telugu or Hindi. No forms to fill.
- **Gathering and drafting done for you** – Gemini searches the manuals and past repairs and drafts a cited guide; the manager approves it and the workers get a printed sheet.
- **Plan before the repair starts** – each guide shows the spare parts in stock and a time estimate.
- **Records stay up to date** – every closed repair is saved without extra typing, so the next breakdown starts from what worked.

Source: ABB, Value of Reliability survey of 3,215 industrial businesses (2023).

Opportunities

How it's different from what exists

	Report by voice or photo	Finds the right past repair	Guide in the crew's language	Nothing invented: parts checked	Tracks if the fix held
CMMS / SAP PM	No, forms	Manual search	No	Not its job	Not flagged
Predictive maintenance	No, sensors	No	No	Not its job	No
Generic AI chatbots	Yes	No, not your data	Unchecked	No, invents parts	No
PlantMemory	**Yes**	**Yes**	**Yes**	**Yes, in code**	**Yes, flags repeats**

How it solves the problem

Typed in by hand	→	Speak, type or photograph the logbook
Answers scattered	→	One search finds the right past repair and manual page
Crew can't read the manual	→	A printed guide in English, Telugu or Hindi
Guided by word of mouth	→	Every step cited; the manager approves before it prints
Time and parts unknown	→	A time estimate and the parts in stock, before work starts
Same breakdown returns	→	Flags repeat faults and fixes that didn't hold

Our USP

Every step cited, every number checked, and every fix tracked to see if it held.

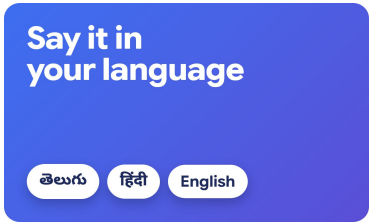
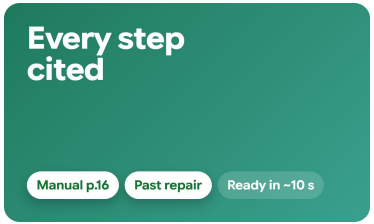
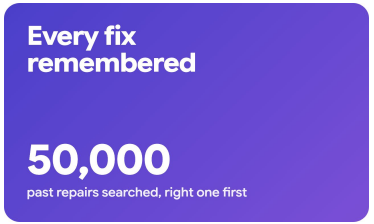
14 / 14
right repair found first among 50,000

333 / 333
planted faults caught by code

0 of 11
false repeat-fault alarms

Figures from PlantMemory's own benchmark and evaluation (14 Oct 2026, gemini-3.1-flash-lite) and scale test (17 Oct 2026, 50,000 real FDA repair reports). Chatbot test: none of the 28 part numbers one gave existed.

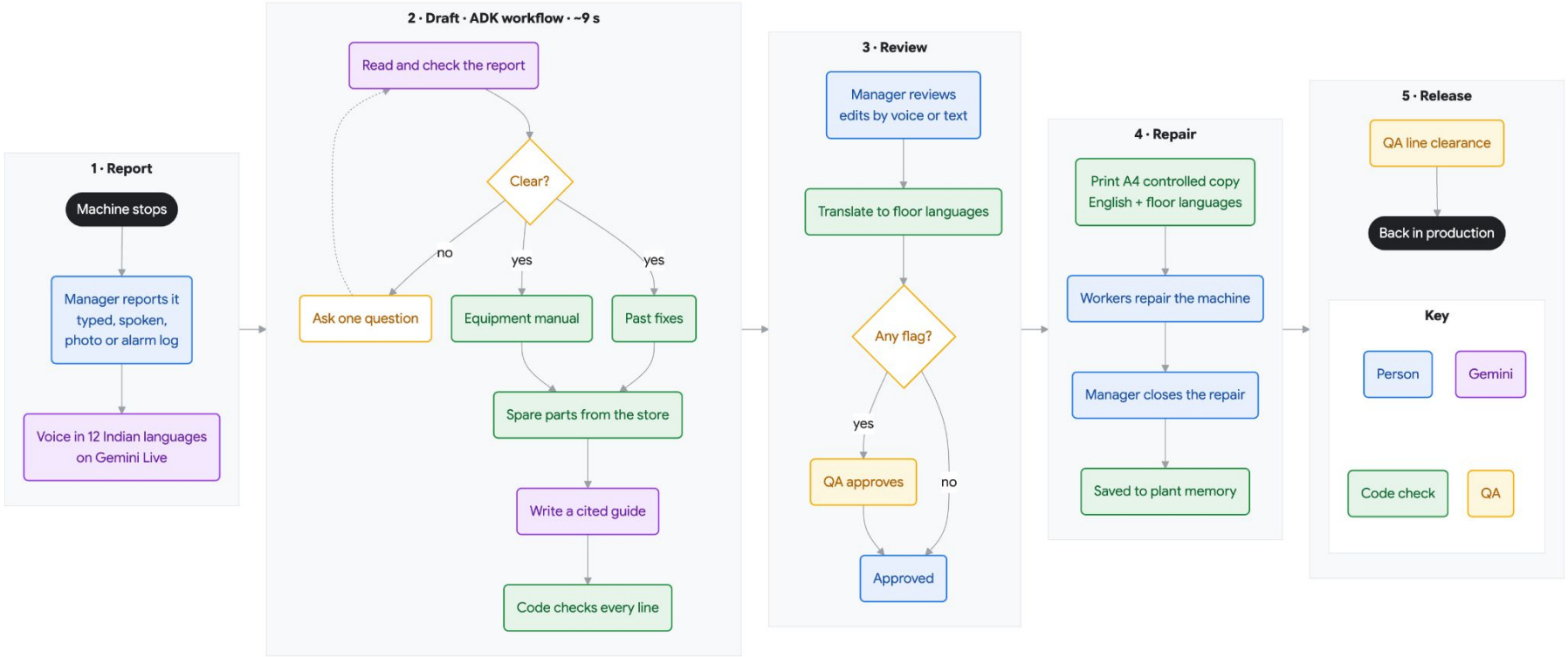
List of features offered by the solution

	Report a breakdown	Read alarm logs	Get a repair guide	Keep the plant's memory
				
What it does	• Type, speak or photograph the logbook • English and many Indian languages, even mixed. • Voice chat asks only for what's missing • One question instead of a guess	• Upload an HMI alarm export or controller log • The error that stopped the machine, with the manual's first actions • Rising warnings flagged before they stop the line	• Steps from the manual and past fixes, each cited • Parts in stock and a time estimate • Manager approves; printed in the crew's languages • Warns when last time's fix didn't hold Print guides in different languages	• Past repairs, manuals by page, parts and stock • Import logbook photos, SAP PM exports, manuals • Search by meaning, across languages
Built with	Gemini Live, Gemini	Log reader in code, manual pages	ADK workflow, Gemini, code checks	Firestore vector search, Gemini embeddings
Outcome	**Reported in seconds, no forms**	**Know what stopped the line**	**A checked fix the crew can follow**	**Every fix makes the next one faster**

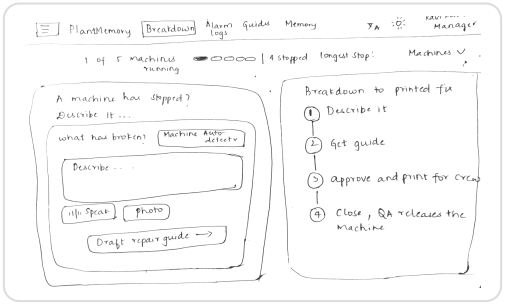
~10 s: median time to a checked guide, 14 Oct 2026. 50,000: scale test on real FDA repair reports, 17 Oct 2026; the first result matched for 14 of 14 test reports. Plant data in the demo is invented.

Process flow diagram

PlantMemory · from a stopped machine to production in five steps

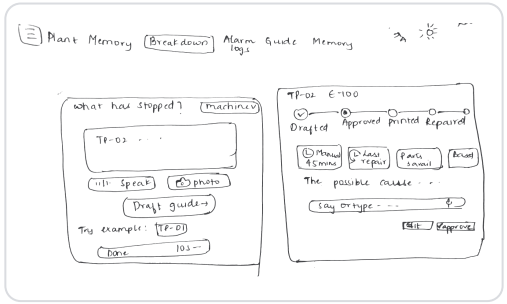


Wireframes/Mock diagrams of the proposed solution



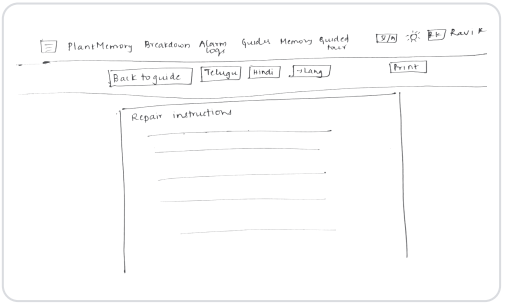
01 Report the breakdown

Type, speak or add a photo. Status on top.



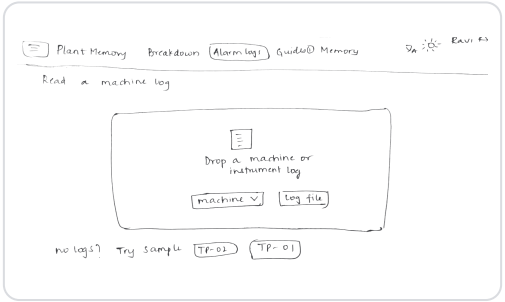
02 Check the draft

Cited steps; edit, then approve.



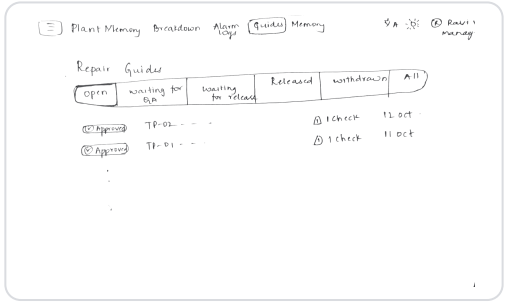
03 Print for the crew

Telugu, Hindi or English, then print.



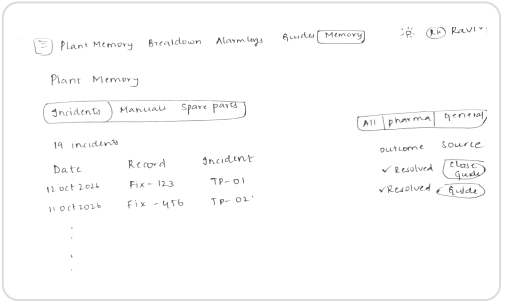
Alarm logs

Drop a machine log to start a report.



Guides

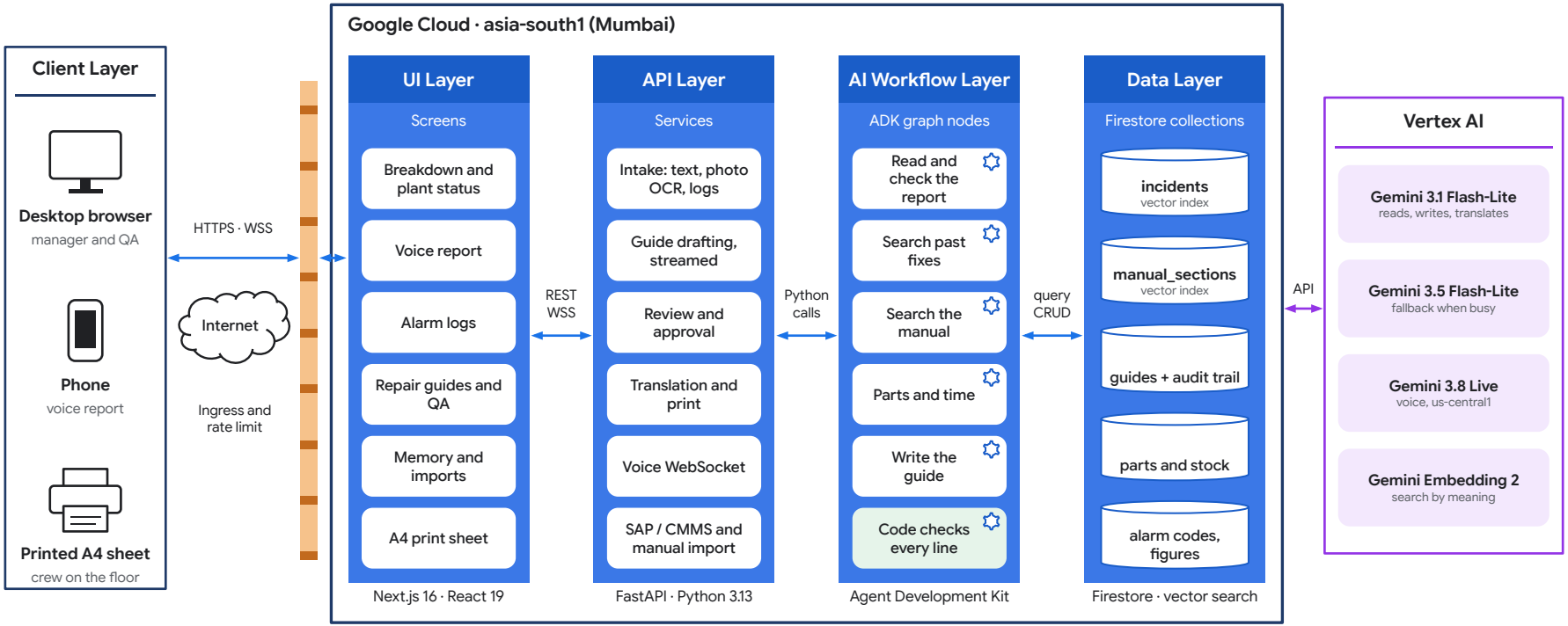
Every guide by stage, open to released.



Memory

Past repairs, manuals and spare parts.

Architecture diagram of the proposed solution



DELIVERY GitHub Actions → Workload Identity → Cloud Build → Artifact Registry → Cloud Run

OPERATIONS Cloud Logging · Error Reporting · Firebase Analytics

Technologies to be used in the solution

01
Gen AI

Gemini 3.1 Flash-Lite
Reads, checks, writes and translates

Gemini 3.5 Flash-Lite
Takes over when the first is busy

Gemini 3.8 Live
Live voice in Indian languages

Gemini Embedding 2
Search by meaning, across languages

Agent Development Kit
Runs the repair guide as a graph workflow

02
Google Cloud

Vertex AI
Every Gemini call, under IAM

Cloud Run
API, UI and voice in one container

Firestore
Memory, vector search, audit trail

Cloud Build · Artifact Registry
Builds and stores every release

Logging · Firebase Analytics
What was used and what broke

03
Application

Python 3.13 · FastAPI · Pydantic
API, a typed model for every guide

Next.js 16 · React 19 · Tailwind 4
The manager's app, phone and desktop

WebSocket · browser audio
Streams voice to Gemini Live

pdfium · Pillow · openpyxl
Manuals, logbook photos, Excel

Print-ready HTML
A4 copy in up to 4 languages

04
Engineering

GitHub Actions
Checks every PR; main deploys itself

Workload Identity Federation
Deploys with no service account keys

pytest · ruff · ESLint
172 automated tests

Docker · uv · pnpm
Locked builds, non-root container

Lighthouse
98 speed, 100 accessibility, phone

WHY THIS STACK

Serverless end to end

Scales to zero when idle

Data stays in asia-south1

No keys to leak or rotate

Estimated implementation cost (optional)

OPERATING COST, YEAR 1

Team salaries 2 engineers, a pharma QA lead and a field trainer	**$39k**	71%
Validation and compliance GAMP 5 validation documents and a security test	**$5k**	9%
Pilots and travel 3 pilot plants: on-site setup, SAP PM / CMMS import	**$4k**	7%
Staff training On-site sessions every shift; videos in Telugu, Hindi, English	**$3.5k**	6%
Outreach Pharma association events and plant demos	**$2k**	4%
Cloud and AI Cloud Run, Firestore, Gemini: $6 per plant a month, measured	**$1.5k**	3%



YEAR 1

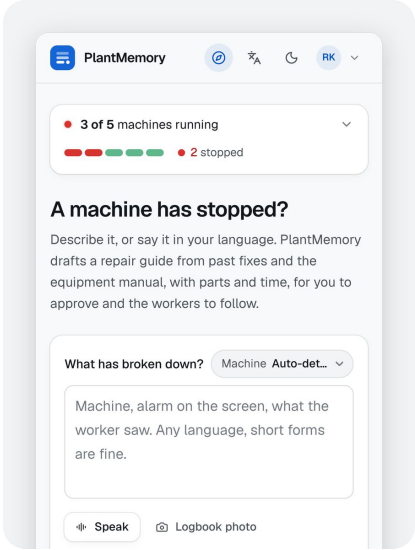
Estimated at **$55,000**

OUR PROTOTYPE, TODAY (MEASURED)

Built for about $800 (8 days, 2 people) · **runs at $6 a plant a month**

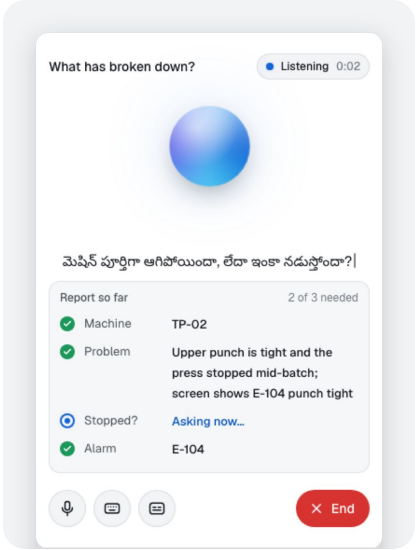
Estimates. Year 1: pilots in 3 plants, growing to 10. Salaries at 2026 Indian market medians (engineer about $12.6k a year). Prototype cost is team time at that rate.

Snapshots of the Prototype



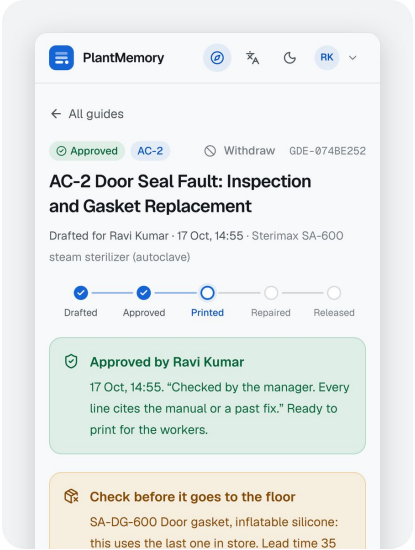
1 Report a breakdown

Type it, speak it, photograph the logbook or upload the alarm log.



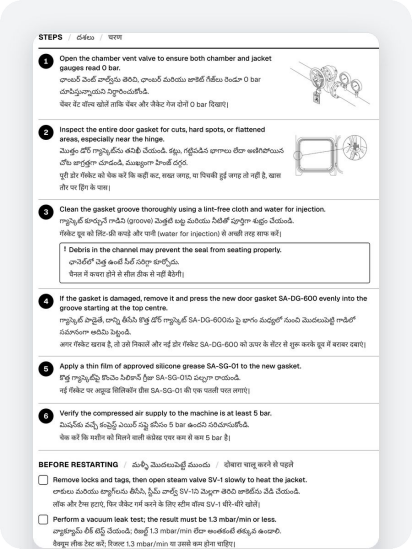
2 Talk it through

A live voice conversation in language of choice, asks only for what's missing.



3 Check the repair guide

Edit the draft with AI, Repeat warnings, the last part in store and the approval track.



4 Print it for the crew

Every step in three languages, with the manual's own figure.

Also in the prototype: alarm-log triage, guides tracked from draft to QA release, logbook and SAP imports, and light and dark themes on phone and desktop.

Prototype Performance report/Benchmarking

9.3 s
median to a checked guide

8 of 8
managers at once, slowest 14.5 s

50,000
past repairs searched in 0.77 s

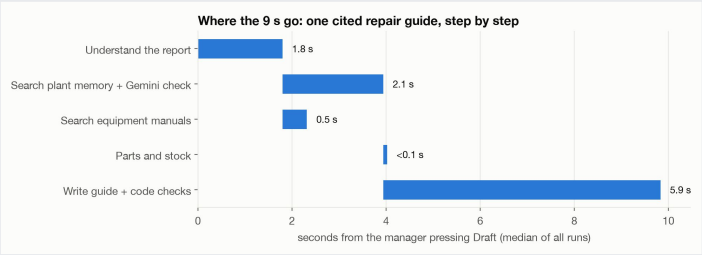
14 / 14
right repair found at 50,000

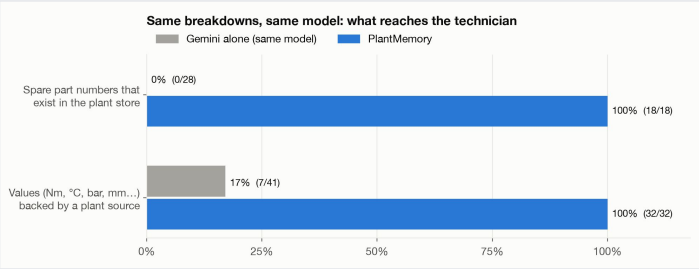
333 / 333
planted faults caught by code

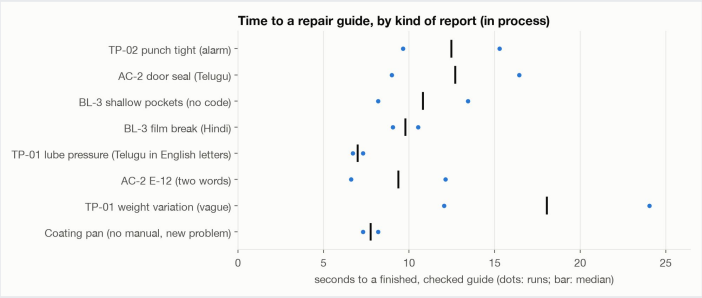
101 / 112
steps cite a manual or past fix

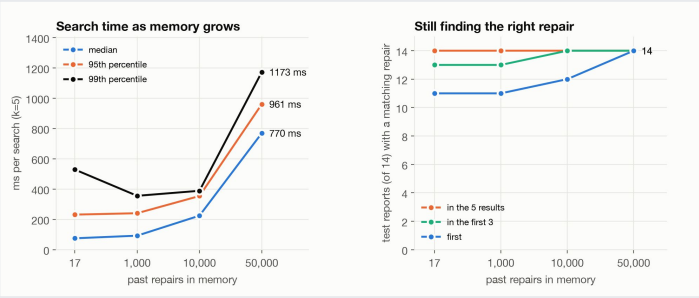
0 / 28
Gemini-alone part numbers that exist

$0.008
Gemini per guide, translated









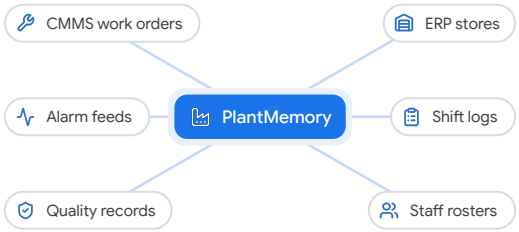
Measured 14 to 17 Oct 2026 with gemini-3.1-flash-lite. Scale test: 50,000 real FDA repair reports in Firestore (Mumbai). Manuals and parts are synthetic; n is small.

Additional Details/Future Development (if any)

CONNECT

Plug into the plant's own systems

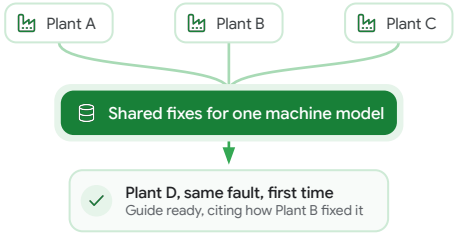
Fewer places to type. Breakdowns, parts and closing records move between the plant's systems with no copying across.



SHARE

One fix, every plant

Plants with the same machine can opt in to pool fixes. When one plant solves a fault, the next one gets the guide on day one.



Opt in. Repair steps are shared; names, stock and people stay in the plant.

LEARN

Pilot in plants, act on feedback

Managers and crews use it on real breakdowns in working plants. Their edits, questions and complaints decide what we build next.



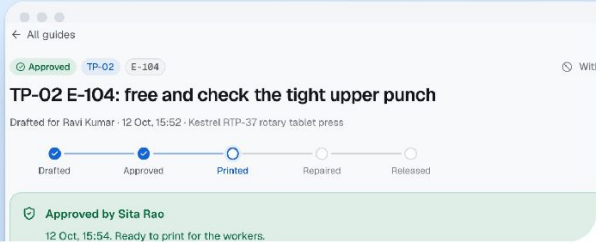
One aim behind all three: take friction out of every input point, so anyone on the floor can report a fault and follow the fix.

Provide links to your:

- GitHub Public Repository - https://github.com/amunjuluri/plantmemory
- Demo Video Link (3 Minutes) - https://www.youtube.com/watch?v=yRuIP2r4Lao
- Final Product Link - https://plantmemory-4l5hbd2imq-el.a.run.app/

Repair guide in 10 seconds

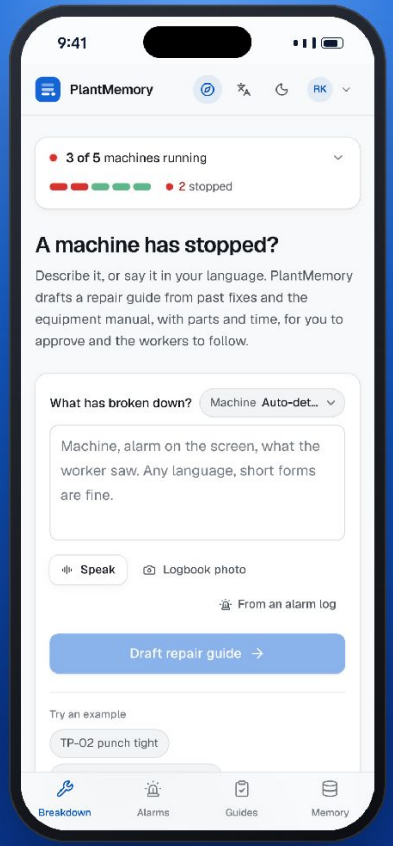
- Warns when last time's fix didn't hold



lost for every hour a line stands still

ABB survey of 3,215 plants

$125k



PlantMemory

Logbooks → repair guide

From scattered logbooks, SAP history and equipment manuals to one checked fix the crew can follow.

Every step cited. Every number checked.

Speak in your language

12 Indian languages, live voice

Printed for the floor

No phones needed on the line

Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.

Thank you

